

Christopher Douglas Robert Wyatt

Curriculum Vitae

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Profile

Bioinformatician specialising in reproducible, scalable pipelines for evolutionary and developmental genomics. I build Nextflow/nf-core workflows for high-throughput sequencing data — RNA-Seq, genomics and proteomics — and currently develop open-source pipelines for evolution and ecology at UCL, where I built **Eco-Flow**, a community working to close “the omics gap in ecology”. Its growing collection spans comparative and ecological genomics, from genome quality assessment and synteny analysis to gene-family evolution and DNA metabarcoding. A fluent programmer (Unix, R, Python, Perl) and an active science communicator and trainer, with high-impact publications stemming from a PhD on alternative splicing at the Centre for Genomic Regulation (CRG, Barcelona).

Education & Research Career

Pipeline Developer

Sequera Labs, Barcelona

June 2021 – Sept 2021

Writing Nextflow pipelines for several startups as well as documenting the training material for Nextflow. This gave me in-depth knowledge of this domain specific language for pipeline development.

Post-doctorate in Evolutionary Genomics / Transcriptomics

Centre for Biodiversity and Environment Research, UCL, London

April 2019 – Feb 2021

Worked on a variety of research projects in the Eusocial Insect Research Group (Seirian Sumner) using machine learning, transcriptomics and proteomics approaches (using Nextflow) to explore wasp caste identity and evolution. Work submitted to bioRxiv prior to publication.

PhD in Biomedicine

Systems Biology, Centre for Genomic Regulation, Barcelona

Sept 2015 – March 2019

Studied the regulation and patterns of alternative splicing during pre-implantation mammalian development.

Selected Projects & Community

Eco-Flow — open-source Nextflow pipelines for evolution & ecology

eco-flow.github.io · github.com/eco-flow

UCL, ongoing

Built **Eco-Flow**, a BBSRC-funded community developing reproducible, nf-core-standard pipelines that close “the omics gap in ecology”. Build and maintain pipelines including:

- **genomeqc** — assess and compare the quality of multiple genome assemblies and their annotations
- **synteny** — synteny analysis across genomes
- **excon** — quantify gene-family expansion and contraction

- DNA metabarcoding (`dietary_amplicons`, `pollen-metabarcoding`, `nanoporem metabarcoding`)
— taxonomic profiling from amplicon and nanopore reads

Training & community: Organise nf-core hackathons and deliver multi-day Nextflow and bioinformatics workshops; an active member of the nf-core community and UCL's Advanced Research Computing (ARC) Centre.

Technical Skills

Languages & tools: Unix/Bash, Perl, R, Python, Matlab

Workflows: Nextflow (nf-core), reproducible & FAIR pipelines

Data types: RNA-Seq, proteomics, genomic and phenotypic data

Publications

1. D. Sampaio Gonçalves, L. Dulac, B. Ritschka, M. Rhinn, A. Klein, **C.D.R. Wyatt**, M. Irimia & W.M. Keyes (2025). Senescent cells exhibit features of developmental signaling centres. *bioRxiv* (preprint). doi:10.1101/2025.10.30.685565
2. T.D. Doyle, E. Jimenez-Guri, J.C. Barnes, C. Hannah, S. Murray, **C.D.R. Wyatt**, O.M. Poole & K.R. Wotton (2025). Long-range pollen transport across the North Sea: Insights from migratory hoverflies landing on a remote oil rig. *Journal of Animal Ecology*, 94(11), 2267-2281. doi:10.1111/1365-2656.70126
3. A. Di Matteo, E. Belloni, D. Pradella, A.M. Chiaravalli, G.M. Pini, M. Bugatti, R. Alfieri, C. Barzan, E. Franganillo Tena, S. Bione, E. Terenzani, F. Sessa, **C.D.R. Wyatt** et al. (2023). Alternative Splicing Changes Promoted by NOVA2 Upregulation in Endothelial Cells and Relevance for Gastric Cancer. *International Journal of Molecular Sciences*, 24(9), 8102. doi:10.3390/ijms24098102
4. **C.D.R. Wyatt**, M.A. Bentley, D. Taylor, E. Favreau, R.E. Brock, B.A. Taylor, E. Bell, E. Leadbeater & S. Sumner (2023). Social complexity, life-history and lineage influence the molecular basis of castes in vespine wasps. *Nature Communications*, 14(1). doi:10.1038/s41467-023-36456-6
5. **C.D.R. Wyatt**, B. Pernaute, A. Gohr, M. Miret-Cuesta, L. Goyeneche, Q. Rovira, M.C. Salzer, E. Boke, O. Bogdanovic, S. Bonnal et al. (2022). A developmentally programmed splicing failure contributes to DNA damage response attenuation during mammalian zygotic genome activation. *Science Advances*, 8(15). doi:10.1126/sciadv.abn4935
6. **C.D.R. Wyatt**, M. Bentley, D. Taylor, R.E. Brock, B.A. Taylor, E. Bell, E. Leadbeater & S. Sumner (2020). Genetic toolkit for sociality predicts castes across the spectrum of social complexity in wasps. *bioRxiv* (preprint). doi:10.1101/2020.12.08.407056
7. E. Belloni, A. Di Matteo, D. Pradella, M. Vacca, **C.D.R. Wyatt**, R. Alfieri, A. Maffia, S. Sabbioneda & C. Ghigna (2019). Gene Expression Profiles Controlled by the Alternative Splicing Factor Nova2 in Endothelial Cells. *Cells*, 8(12), 1498. doi:10.3390/cells8121498
8. R.J. Southon, E.F. Bell, P. Graystock, **C.D.R. Wyatt**, A.N. Radford & S. Sumner (2019). High indirect fitness benefits for helpers across the nesting cycle in the tropical paper wasp *Polistes canadensis*. *Molecular Ecology*, 28(13), 3271-3284. doi:10.1111/mec.15137
9. F. Marlétaz, P.N. Firbas, I. Maeso, J.J. Tena, O. Bogdanovic, M. Perry, **C.D.R. Wyatt**, E. de la Calle-Mustienes, S. Bertrand, D. Burguera et al. (2018). Amphioxus functional genomics and the origins of vertebrate gene regulation. *Nature*, 564(7734), 64-70. doi:10.1038/s41586-018-0734-6

10. P. Kennedy, G. Baron, B. Qiu, D. Freitak, H. Helanterä, E.R. Hunt, F. Manfredini, T. O'Shea-Wheller, S. Patalano, C.D. Pull, T. Sasaki, D. Taylor, **C.D.R. Wyatt** et al. (2017). Deconstructing Superorganisms and Societies to Address Big Questions in Biology. *Trends in Ecology & Evolution*, 32(11), 861-872. doi:10.1016/j.tree.2017.08.004
11. I. Maeso, T.L. Dunwell, **C.D.R. Wyatt**, F. Marlétaz, B. Vető, J.A. Bernal, S. Quah, M. Irimia & P.W.H. Holland (2016). Evolutionary origin and functional divergence of totipotent cell homeobox genes in eutherian mammals. *BMC Biology*, 14(1). doi:10.1186/s12915-016-0267-0
12. M. Guillemin, L. Contreras-Porcia, M.E. Ramírez, E.C. Macaya, C.B. Contador, H. Woods, **C. Wyatt** & J. Brodie (2016). The bladed Bangiales (Rhodophyta) of the South Eastern Pacific: Molecular species delimitation reveals extensive diversity. *Molecular Phylogenetics and Evolution*, 94, 814-826. doi:10.1016/j.ympev.2015.09.027
13. S. Patalano, A. Vlasova, **C. Wyatt**, P. Ewels, F. Camara, P.G. Ferreira, C.L. Asher, T.P. Jurkowski, A. Segonds-Pichon, M. Bachman et al. (2015). Molecular signatures of plastic phenotypes in two eusocial insect species with simple societies. *Proceedings of the National Academy of Sciences*, 112(45), 13970-13975. doi:10.1073/pnas.1515937112
14. C. Giampietro, G. Deflorian, S. Gallo, A. Di Matteo, D. Pradella, S. Bonomi, E. Belloni, D. Nyqvist, V. Quaranta, S. Confalonieri, G. Bertalot, F. Orsenigo, F. Pisati, E. Ferrero, G. Biamonti, E. Fredrickx, C. Taveggia, **C.D.R. Wyatt** et al. (2015). The alternative splicing factor Nova2 regulates vascular development and lumen formation. *Nature Communications*, 6(1). doi:10.1038/ncomms9479